WHICH BOUNDEDNESS SETS CAN OCCUR?

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AI assistance was used in adapting and checking the argument and the exposition.

After stating Theorem 1.1, Andersson writes: "We are unaware of whether this completely characterizes all possible sets A, and find it an interesting question to determine precisely what sets A can occur." See §1 of "Exotic operators on the Hardy spaces of the infinite-dimensional torus." The inference is to characterize the exact boundedness sets among the theorem's common-domain operators.

Here $A = \{p \in [1,\infty] : T \text{ extends boundedly on } H^p(\mathbb{T}^\infty)\}$, and Theorem 1.1 constructs every such set whose finite part is relatively closed, using one domain dense at every finite exponent and weak-$*$ dense at infinity.

The external block construction used in the proof is Andersson, *Exotic operators on the Hardy spaces of the infinite-dimensional torus*, Lemmas 2.7–2.8 and the proof of Theorem 1.1, arXiv:2609.12859.

Answered: exact characterization. Evidence: Sketch.

Exactly the sets $A \subseteq [1,\infty]$ for which $A \setminus \{\infty\}$ is F_σ in $[1,\infty)$.

Proof. Let D be the common dense domain, with D and $T(D)$ contained in every finite H^p. For each nonzero $f \in D$, the ratio $r_f(p) = \|Tf\|_p/\|f\|_p$ is continuous: on a compact interval $a \le p \le b$, $|g|^p \le |g|^a + |g|^b$ gives dominated convergence for each $g \in \cap_{q<\infty} H^q$. Thus $N(p) = \sup_{f\neq 0} r_f(p)$ is lower semicontinuous. Boundedness is equivalent to $N(p) < \infty$, so the finite part of A is $\bigcup_{k\ge 1}\{N \le k\}$, an F_σ set.

Conversely, write $A \cap [1,\infty) = \bigcup_k F_k$ with F_k increasing and closed, and put $U_k = [1,\infty)\setminus F_k$. Cover each U_k by compact intervals inside relatively open bounded intervals with overlap at most three. Andersson's Lemmas 2.7–2.8 in arXiv:2609.12859 supply, for each compact interval, an operator R_ν of norm at most one outside its open interval and norm at least $1+\rho_\nu$ on a neighborhood of the compact interval. Taking an ℓ_ν-fold tensor block on fresh variables gives norm $\|R_\nu\|_p^{\ell_\nu}$. Choose ℓ_ν so $2^{-\nu}(1+\rho_\nu)^{\ell_\nu} \ge k(\nu)$ and sum $T = \sum_\nu 2^{-\nu} R_\nu^{(\ell_\nu)}$ on analytic polynomials. Each polynomial uses only finitely many blocks. At $p \in A$, only finitely many blocks can have norm above $2^{-\nu}$; for $p \notin A$, a block at every level forces the polynomial-domain operator norm to be arbitrarily large. If $\infty \notin A$, add Andersson's one-variable Volterra operator, bounded at every finite exponent and unbounded on H^∞, acting in a fresh variable. This realizes the prescribed finite part and the chosen behavior at infinity. □

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